FES | CONSILIUM'26 THE ANNUAL BUSINESS CONCLAVE OF NSUT | NSUT NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY

MAUDRIK V7.0

POLICY CASE COMPETITION

PROBLEM STATEMENT

BACKGROUND

1) India's examination system is one of the largest mechanisms through which educational and employment opportunities are allocated.

2) From university entrance examinations to public sector recruitment, millions of candidates compete each year for opportunities that can determine their educational and professional trajectories.

3) However, the scale and stakes of the system have created a parallel challenge: organised examination malpractice.

4) Paper leaks, impersonation, insider collusion, hacking, manipulation of examination centres, unauthorised access to question papers and organised cheating networks have demonstrated that examination security is no longer merely an administrative issue.

5) It is now a question of:
Institutional Integrity | Cybersecurity | Public Trust | State Capacity

6) A compromised examination imposes significant costs on candidates, governments and institutions while weakening confidence in merit-based selection.



CURRENT SITUATION

India has progressively introduced technology and institutional controls across the examination ecosystem.

EXISTING MEASURES

1. Computer-based examinations
2. Biometric authentication
3. CCTV and centre surveillance
4. Encrypted examination material
5. Digital Reg. and verification
6. AI-Assistes monitoring
7. Centralised examination platform
8. Secure printing and logistics
9. Cybersecurity protocols

However, greater technological sophistication does not automatically eliminate malpractice. It can simply shift the point of vulnerability.

A physical examination may be vulnerable to paper leakage or insider intervention.

A digital examination may become vulnerable to cyberattacks, compromised credentials, insider access or technological failure

CHALLENGES AND POLICY ISSUES

The examination ecosystem involves multiple stakeholders:

Examination Authorities | Governments | Technology Providers | Logistics Agencies | Examination Centres | Invigilators | Candidates | Coaching Institutions | Cybersecurity Firms | Law Enforcement

Each additional layer can introduce another point of vulnerability.

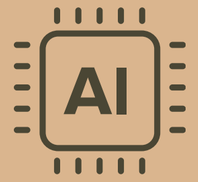
Key Challenges:

1) Insider risk	**2) Cyber-physical risk**	**3) Centralisation Risk**	**4) Third party dependence**	**5) Cost vs. Security**	**6) Privacy vs Surveillance**

Fundamental Challenge

Security must be strengthened without making the examination ecosystem prohibitively expensive, inaccessible or overly centralised.

STRATEGIC OPTIONS AND OPPORTUNITIES



AI-Assisted Security



Multi-layered Authentication



Zero-trust architecture



Risk based security



Integrated Governance



Distributed Question Paper Architecture

The objective is not to create a system that is impossible to compromise.
It is to create one where the probability of compromise is reduced, the impact of failure is contained and the cost of attack becomes significantly higher than its expected benefit.

PROBLEM STATEMENT

Designing India's National Examination Security Architecture

The Government of India has asked you to redesign the security architecture of India's high-stakes examination ecosystem. The objective is to create a comprehensive and secure framework that protects the examination process from end to end, while ensuring confidentiality, integrity, accountability, and fairness at every stage.

Your framework must address the entire chain of the examination lifecycle, identifying potential security vulnerabilities, access risks, points of failure, and mechanisms required to prevent, detect, and respond to threats across each stage:

- Question Creation
- Question Storage
- Personnel Access
- Transmission
- Examination Centre
- Candidate Authentication
- Examination
- Evaluation & Results

The proposed architecture should therefore provide security across the complete examination ecosystem rather than focusing on any single stage of the process.



YOUR TASK

Design a 10-Year National Examination Security Architecture for India

Your proposal must address:

1. Should examination security be centralised, decentralised or hybrid?
2. How can India reduce insider, physical, cyber and operational risks?
3. What role should AI, biometrics, encryption and surveillance play?
4. How should responsibility, audits and liability be assigned after a breach?
5. What level of security spending is economically justified?
6. How can security improve without compromising accessibility and affordability?

Your proposal must include:

1. RISK FRAMEWORK – Prioritise key physical, cyber, insider and operational risks.
2. GOVERNANCE MODEL – Define roles of authorities, governments, providers and stakeholders.
3. TECHNOLOGY FRAMEWORK – Recommend deployment of AI, biometrics, encryption and surveillance.
4. ACCOUNTABILITY – Establish clear responsibility, audits and liability mechanisms.
5. 10-YEAR ROADMAP – Present a phased implementation plan.
6. COST-BENEFIT – Estimate costs and justify the proposed expenditure.

Show why your model is more resilient than India's fragmented approach and highlight its key trade-offs.

Rules and submission

1) AI tools may be used for research/reference only. The final solution and deck must be independently created by the team. **AI-generated/LLM-made decks will be disqualified.**

2) Research & Sources: Internet research is allowed. **Cite sources for key data, facts, and insights used.**

3) **Submissions must be original.** Plagiarism or fabricated data will lead to disqualification.

4) **Maximum 10 slides**, including Intro/Appendix/Backup slides.

5) Reasonable assumptions are allowed where reliable data is unavailable, but must be stated clearly.

6) Submit in PDF format. **File name: TeamName.pdf**

7) Final round (offline): 8 mins presentation + 4 mins Q&A.

8) Organizers' and jury's decision shall be final.

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